WEB 2

Dear friends just pray that whoever made these notes becomes a CEO someday 🤌

Practical XML program codes

20. Create a simple XML document to display the address book.

address.xml

```
<?xml version="1.0" encoding="UTF-8"?>

<addressbook>

   <contact>
      <name>Arun Kumar</name>
      <phone>9876543210</phone>
      <email>arun@gmail.com</email>
      <address>Chennai</address>
   </contact>

</addressbook>
```

21. Create a simple XML document to display the customer details.

Customer21.xml

```
<?xml version="1.0" encoding="UTF-8"?>

<customers>

   <customer>
      <id>103</id>
      <name>Priya</name>
      <phone>9876512345</phone>
      <email>priya@gmail.com</email>
      <city>Madurai</city>
```

```
    </customer>

</customers>
```

22. Create an internal DTD and validate the XML document for the customer details.

customer21.xml

```
<?xml version="1.0"?>

<!DOCTYPE customers [
    <!ELEMENT customers (customer+)>
    <!ELEMENT customer (id,name,phone,email,city)>
    <!ELEMENT id (#PCDATA)>
    <!ELEMENT name (#PCDATA)>
    <!ELEMENT phone (#PCDATA)>
    <!ELEMENT email (#PCDATA)>
    <!ELEMENT city (#PCDATA)>
]>

<customers>
    <customer>
        <id>101</id>
        <name>Arun</name>
        <phone>9876543210</phone>
        <email>arun@gmail.com</email>
        <city>Chennai</city>
    </customer>
</customers>
```

23. Create an external DTD and validate the XML document with the customer details.

Customer.dtd

```
<!ELEMENT customers (customer+)>
```

```
<!ELEMENT customer (id,name,phone,email,city)>
<!ELEMENT id (#PCDATA)>
<!ELEMENT name (#PCDATA)>
<!ELEMENT phone (#PCDATA)>
<!ELEMENT email (#PCDATA)>
<!ELEMENT city (#PCDATA)>
```

Customer.xml

```
<?xml version="1.0"?>

<!DOCTYPE customers SYSTEM "customer.dtd">

<customers>
   <customer>
      <id>101</id>
      <name>Arun</name>
      <phone>9876543210</phone>
      <email>arun@gmail.com</email>
      <city>Chennai</city>
   </customer>
</customers>
```

24. Define the XML structure containing person details and create an XSL (Extensible Stylesheet Language) document to transform. Display the data in HTML table format.

person.xml

```
<?xml version="1.0"?>

<?xml-stylesheet type="text/xsl" href="person.xsl"?>

<persons>
   <person>
      <name>Arun</name>
      <age>21</age>
      <city>Chennai</city>
```

```
    </person>

    <person>
        <name>Rahul</name>
        <age>22</age>
        <city>Madurai</city>
    </person>

    <person>
        <name>Priya</name>
        <age>20</age>
        <city>Coimbatore</city>
    </person>
</persons>
```

person.xsl

```
<?xml version="1.0"?>
<xsl:stylesheet version="1.0"
xmlns:xsl="http://www.w3.org/1999/XSL/Transform">

<xsl:output method="html"/>

<xsl:template match="/">
<html>
<body>
<h2>Person Details</h2>
<table border="1">
<tr><th>Name</th><th>Age</th><th>City</th></tr>
<xsl:for-each select="persons/person">
<tr>
<td><xsl:value-of select="name"/></td>
<td><xsl:value-of select="age"/></td>
<td><xsl:value-of select="city"/></td>
</tr>
</xsl:for-each>
</table>
</body>
```

```
</html>
</xsl:template>

</xsl:stylesheet>
```

Expected output

Name	Age	City
Arun	21	Chennai
Rahul	22	Madurai
Priya	20	Coimbatore

25. Write an XML file which will display the Book information which includes the following:

(a)(i) Title of the book
(ii) Author Name
(iii) ISBN number
(iv) Publisher name
(v) Edition
(vi) Price
b. Write an External Document Type Definition (DTD) to validate the above XML file.
c. Display the XML file in HTML table format.

book.xml

```
<?xml version="1.0"?>
<?xml-stylesheet type="text/xsl" href="book.xsl"?>
<!DOCTYPE books SYSTEM "book.dtd">
<books>
<book>
<title>Web Technology</title>
<author>John</author>
<isbn>1234567890</isbn>
<publisher>ABC Publications</publisher>
<edition>2nd</edition>
<price>500</price>
</book>
</books>
```

book.xsl

```
<?xml version="1.0"?>
<xsl:stylesheet version="1.0"
xmlns:xsl="http://www.w3.org/1999/XSL/Transform">
<xsl:template match="/">
<html><body>
<h2>Book Information</h2>
<table border="1">
<tr><th>Title</th><th>Author</th><th>ISBN</th><th>Publisher</th><th>Edition</
th><th>Price</th></tr>
<xsl:for-each select="books/book">
<tr>
<td><xsl:value-of select="title"/></td>
<td><xsl:value-of select="author"/></td>
<td><xsl:value-of select="isbn"/></td>
<td><xsl:value-of select="publisher"/></td>
<td><xsl:value-of select="edition"/></td>
<td><xsl:value-of select="price"/></td>
</tr>
</xsl:for-each>
</table>
</body></html>
</xsl:template>
</xsl:stylesheet>
```

book.dtd

```
<!ELEMENT books (book+)>
<!ELEMENT book (title,author,isbn,publisher,edition,price)>
<!ELEMENT title (#PCDATA)>
<!ELEMENT author (#PCDATA)>
<!ELEMENT isbn (#PCDATA)>
<!ELEMENT publisher (#PCDATA)>
<!ELEMENT edition (#PCDATA)>
<!ELEMENT price (#PCDATA)>
```

Expected output

Book Information

Title	Author	ISBN	Publisher	Edition	Price
Web Technology	John	1234567890	ABC Publications	2nd	500